Software Architecture Document (SAD)
Physiological Digital Twin Berbasis CAPAR

Dokumen Software Architecture Document (SAD) ini mendefinisikan arsitektur perangkat lunak Physiological Digital Twin berbasis Context-Aware Personalized Autonomic Recovery (CAPAR). Arsitektur dirancang menggunakan prinsip reference software architecture, traceability, modularity, explainability, dan governance.

1. Tujuan Arsitektur

Arsitektur bertujuan menyediakan fondasi perangkat lunak untuk:

- mengintegrasikan wearable observation, clinical observation, dan contextual input;
- melakukan physiological state estimation;
- membangun resilience status;
- menghasilkan CAPAR event-state;
- menyediakan explainable decision support.

Arsitektur tidak hanya berorientasi anomaly detection, tetapi representasi dinamis kondisi fisiologis individu.

2. Architectural Drivers

Driver utama:

1. Physiological Meaning
Setiap output harus memiliki interpretasi fisiologis.

2. Personalization
Model harus beradaptasi terhadap baseline individu.

3. Context Awareness
Respons harus dibandingkan dengan konteks aktivitas.

4. Explainability
Setiap keputusan harus memiliki evidence.

5. Longitudinal Modeling
Sistem harus menangani trajectory, persistence, recovery, dan relapse.

6. Governance
Data, model, threshold, dan keputusan harus dapat ditelusuri.

3. Architectural Viewpoint

Berdasarkan pendekatan ISO/IEC/IEEE 42010:

Stakeholder:
- Physiologist
- Clinician
- Researcher
- Software Engineer
- Data Scientist
- Participant

Architecture concerns:
- accuracy
- reliability
- interpretability
- scalability
- privacy
- maintainability
- validation

4. High Level Architecture

```
      User / Clinician

          |
          v

   Application & Visualization Layer
   - Dashboard
   - Trajectory Viewer
   - XAI Interface
   - Report Generator

          |
          v

   Physiological Intelligence Layer
   - CAPAR Engine
   - State Estimator
   - Resilience Engine
   - Phenotype Classifier

          |
          v
```

Evidence Processing Layer
- Feature Extraction
- Baseline Engine
- Context Engine
- Quality Engine

|
v

Data Acquisition Layer
- Wearable
- Clinical Data
- EMA / Context

5. Architectural Layers

Layer 1: Multimodal Acquisition Layer

Input:
- HR
- RR/IBI
- HRV
- DFA
- ACC
- Sleep
- Medication
- Clinical parameters

Layer 2: Governed Information Layer

Function:
- validation
- normalization
- provenance
- quality score
- privacy control

Layer 3: Analytical Evidence Layer

Function:
- feature generation
- baseline modeling
- state estimation
- CAPAR inference

Layer 4: Operationalization Layer

Function:
- visualization
- decision support
- XAI
- reporting

6. Core Components

Component 1:
Data Ingestion Service

Component 2:
Signal Processing Service

Component 3:
Context Interpretation Service

Component 4:
Personal Baseline Service

Component 5:
CAPAR State Engine

Component 6:
Physiological Digital Twin Engine

Component 7:
Explanation Engine

Component 8:
Application Service

7. Component Interaction

```
Wearable
  |
  v
Data Ingestion
  |
  v
```

```
Quality Engine
  |
  v
Feature Engine
  |
  +----------------+
  |                |
Context Engine   Clinical Engine
  |                |
  +----------------+
         |
         v
     CAPAR Engine
         |
         v
 Digital Twin State
         |
         v
 XAI + Decision Support
```

8. Data Architecture

Data entities:

Participant Entity
- identity
- demographic

Clinical Entity
- risk factor
- examination
- laboratory

Observation Entity
- timestamp
- HR
- RR
- HRV
- sensor quality

Context Entity
- activity
- sleep

- stress
- medication

Episode Entity
- onset
- peak
- duration
- recovery
- relapse

Twin State Entity
- resilience state
- latent variable
- confidence

Explanation Entity
- evidence
- uncertainty

9. Dynamic Model Architecture

State model:

$u(k)$:
activity, context, stress, sleep, medication

$d(k)$:
noise, unobserved load, environment

$c(k)$:
risk factor, examination, laboratory

$z(k)$:
latent physiological state

$X(k)$:
resilience state

$s(k)$:
CAPAR event-state

Equations:

$z(k+1)=f(z(k),u(k),d(k))+w(k)$

$y(k)=h(z(k),u(k))+v(k)$

$X(k)=g(z(k),c(k),Bpersonal,e(k))$

$s(k+1)=T(s(k),D(k),P(k),R(k))$

10. Variability and Extension Points

Common core:

- CAPAR engine
- state model
- data governance
- XAI framework

Variation points:

- sensor type
- clinical dataset
- feature selection
- inference algorithm
- visualization platform
- deployment environment

Extension:

- additional biomarkers
- additional physiological models
- machine learning models

11. Quality Attribute Design

Performance:
- real-time state update

Reliability:
- robust handling missing data

Security:

- encryption and access control

Maintainability:
- modular component design

Explainability:
- evidence-based output

Interoperability:
- standard data exchange

12. Deployment Architecture

Edge Layer:
- wearable communication
- initial filtering

Cloud Layer:
- database
- inference engine
- model management

Application Layer:
- web dashboard
- mobile application
- research interface

13. Architectural Governance

Governance rules:

1. Clinical interpretation tidak boleh berasal dari black-box output saja.

2. Threshold harus memiliki sumber:
   - literature-based
   - empirical calibration

3. Model version harus dicatat.

4. Semua output memiliki provenance.

5. Human oversight tetap diperlukan.

14. Architecture Evaluation Plan

Evaluasi arsitektur:

Scenario 1:
Continuous monitoring.

Scenario 2:
Recovery trajectory estimation.

Scenario 3:
Relapse detection.

Scenario 4:
Clinical-context fusion.

Metrics:
- latency
- F1
- calibration
- stability
- explanation validity

15. Kesimpulan

Software Architecture Document ini mendefinisikan struktur arsitektur CAPAR Digital Twin sebagai sistem multimodal, context-aware, explainable, dan longitudinal. Arsitektur ini menjadi penghubung antara konsep fisiologi, kebutuhan perangkat lunak, dan implementasi engineering.